



Manipulation Aware Refusal Mechanism

Team Members

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Problem Statement

Modern LLMs can be **psychologically manipulated through conversation** (e.g., guilt, persuasion, intimidation) to produce unsafe or unintended outputs. Our goal is to make the model **recognize and resist such manipulation**, without becoming unhelpful or rude.

Datasets:

- **CON** → label is assigned **only if annotators agree**
- **MAJ** → label is assigned by **majority voting**

Manipulation types includes:

- Denial, Evasion, Feigning Innocence, Rationalization, Playing the Victim Role, Playing the Servant Role, Shaming or Belittlement, Intimidation, Brandishing Anger, Accusation, Persuasion or Seduction

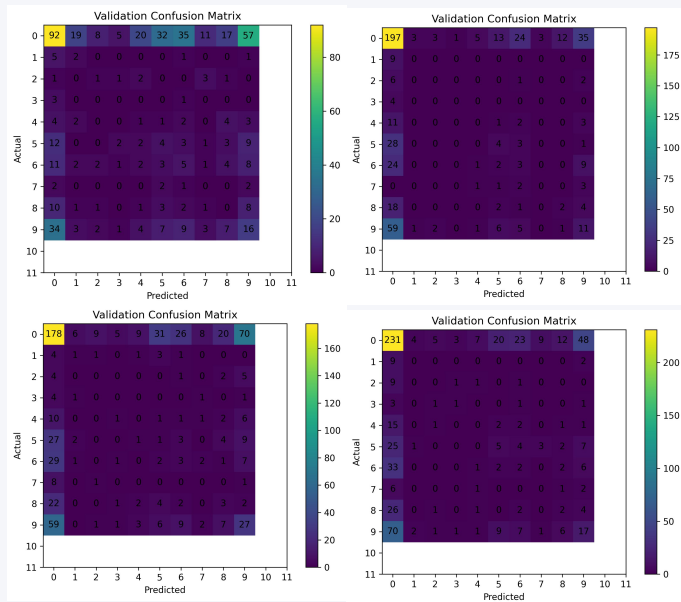
Tasks

1. **Manipulation Detection**
 - Input: Partial conversation
 - Output: Manipulation Technique if there is Manipulation, otherwise No manipulation
2. **Response Adaptation**
 - Use detection signal to:
 - Avoid being manipulated
 - Maintain helpful, polite responses

Note: In this presentation we will focus on **Task-1**

Baseline Models & Validation Behavior

Evaluation of LSTM and BiLSTM models across CON and MAJ dataset settings reveals critical performance gaps.



22.8%

LSTM CON Accuracy

Macro F1: ~0.10

40.9%

BiLSTM CON Accuracy

Macro F1: ~0.10

31.9%

LSTM MAJ Accuracy

Macro F1: ~0.10

38.6%

BiLSTM MAJ Accuracy

Macro F1: ~0.11

Large Accuracy-F1 Gap

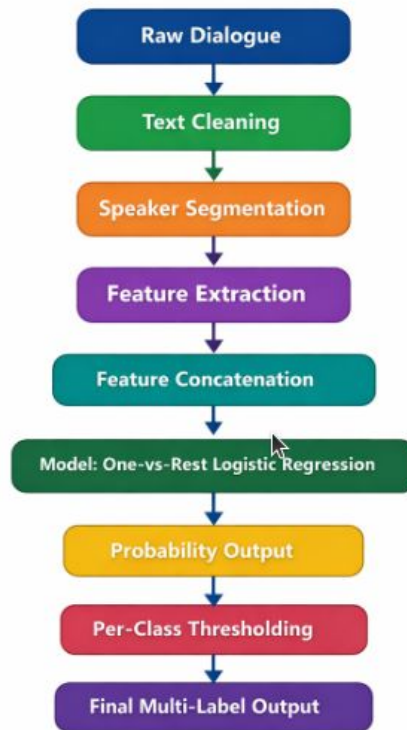
Models heavily biased toward dominant classes, ignoring minority categories

No Real Understanding

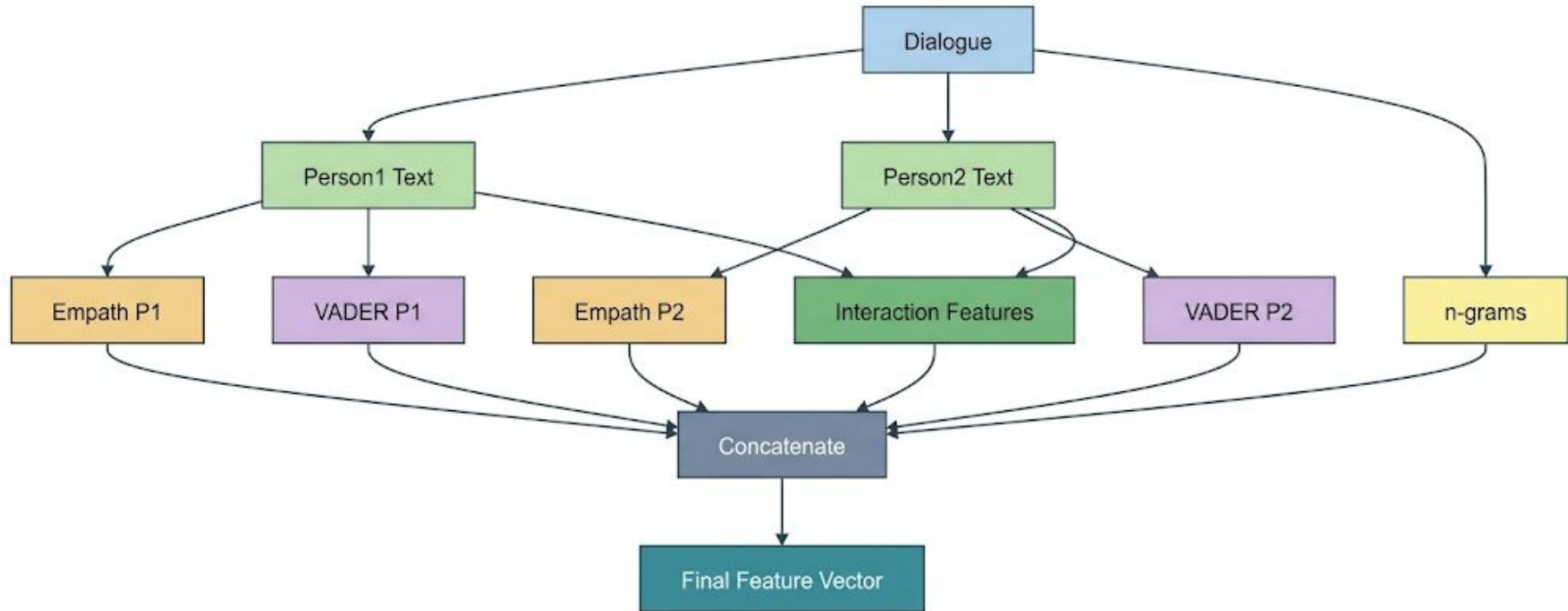
Improved accuracy without capturing fine-grained manipulation categories

Rule Based transformer models

- Dialogue segmented into Person1/Person2 via regex parsing for speaker-wise modeling
- Empath features → semantic categories (anger, power, deception, etc.)
- VADER → sentiment per speaker (pos, neg, neu, compound)
- Interaction features → pronoun ratios, dominance, question counts
- Concatenated feature vector → One-vs-Rest Logistic Regression + per-class thresholding



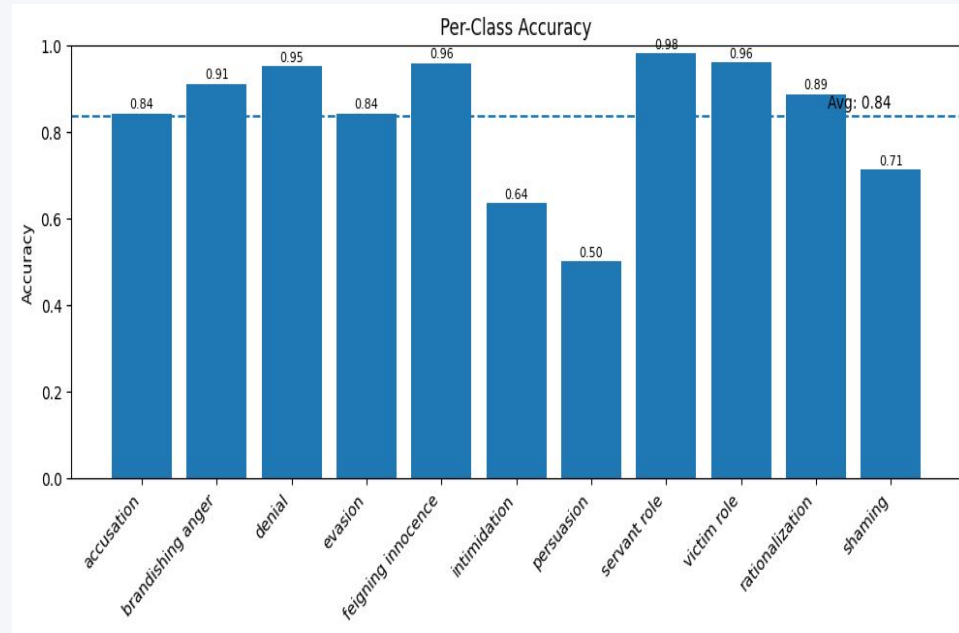
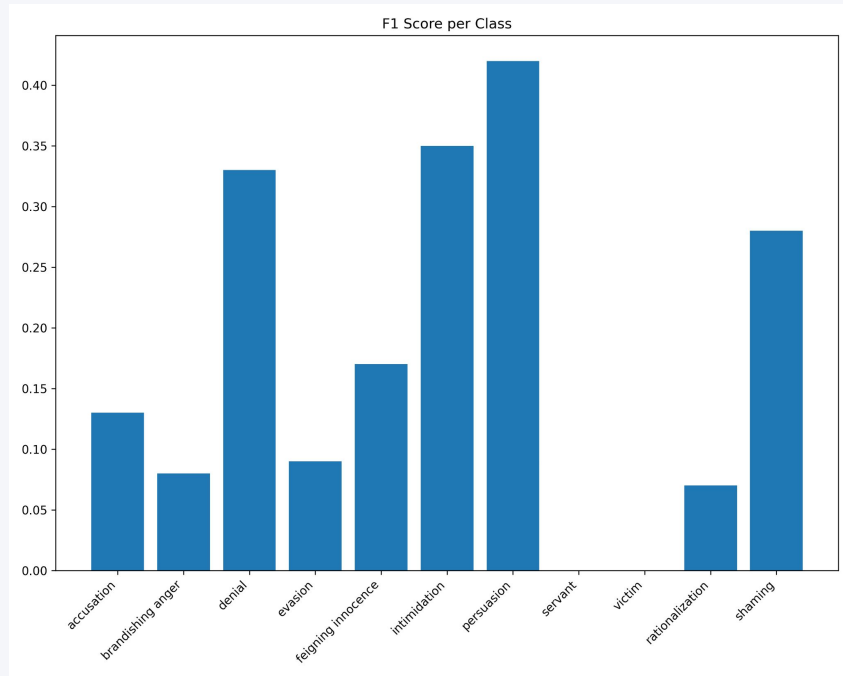
Feature Extraction Pipeline



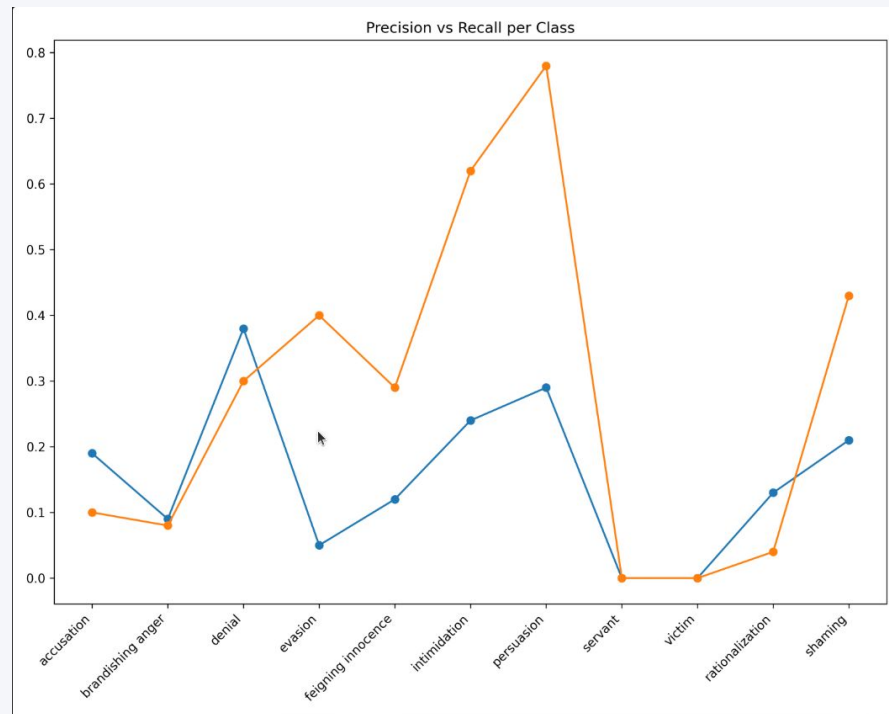
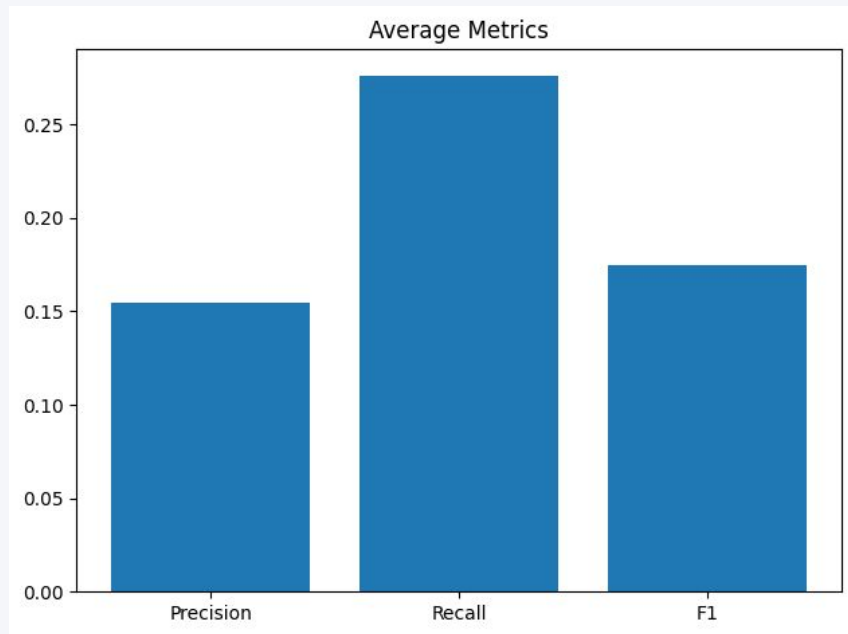
Results

- Average accuracy is 83.6% over all manipulation labels
- Strong Classes:
 - Persuasion/Seduction ($F1 \approx 0.42$)
 - Intimidation (high recall ≈ 0.62)
 - Shaming ($F1 \approx 0.28$)
- Observations:
 - Model captures high-frequency, pattern-driven techniques well
 - Good recall indicates effective detection of manipulation signals
- Challenges:
 - Low performance on rare classes (e.g., victim/servant roles)
 - Data imbalance and subtle language reduce precision
- Key Takeaway:
 - Feature-driven approach successfully captures major manipulation patterns, but performance is limited by class imbalance and representation capacity

Plots



Plots



Intent-Aware Prompting (IAP) Pipeline

Detecting Psychological Manipulation in Dialogues via LLM-Driven Intent Analysis

Core Idea: Separate "what is said" (surface dialogue) from "why it is said" (latent intent) to improve manipulation detection with reduced false negatives.

Architecture & Pipeline Flow

STEP 1

Data Loading

`data_loader.py`

- ▶ Parse MentalManip CSV dataset
- ▶ Extract dialogue text & ground-truth labels
- ▶ Handle technique & vulnerability annotations

STEP 2

Intent Extraction

`intent_extractor.py`

- ▶ Parallel extraction for Person A & Person B
- ▶ LLM acts as psycholinguist (Theory of Mind)
- ▶ Identifies guilt-tripping, gaslighting, blame shifting, emotional blackmail, etc.

STEP 3

Classification

`classifier.py`

- ▶ Combines dialogue + both intent vectors
- ▶ LLM outputs JSON: manipulative (0/1), confidence, technique, explanation
- ▶ Results saved to predictions.csv

Supporting Modules: `llm_client.py` (unified API layer) · `prompts.py` (prompt templates) · `config.py` (settings) · `evaluation.py` (metrics) · `pipeline.py` (async orchestrator)

Key Design Decisions & Technical Details



Multi-Provider LLM Abstraction

- ▶ Unified `call_llm()` dispatcher for OpenAI, Anthropic & Groq
- ▶ Groq uses OpenAI SDK with custom `base_url`
- ▶ Exponential backoff retry for rate limits (2→3→5→9s)
- ▶ Configurable temperature & max tokens per step



Prompt Engineering Strategy

- ▶ System prompt: LLM = expert psycholinguist role
- ▶ Intent extraction focuses on latent manipulation tactics
- ▶ Classification combines dialogue + dual intent vectors
- ▶ Structured JSON output: label, confidence, technique



Async Concurrency Model

- ▶ asyncio-based pipeline with configurable semaphore
- ▶ Parallel intent extraction for Person A & B
- ▶ Concurrent dialogue processing (default: 5 workers)
- ▶ Non-blocking API calls maximize throughput



Evaluation & Safety Focus

- ▶ Key metric: False Negative Rate (FNR) reduction
- ▶ Missed manipulation = safety risk → prioritize recall
- ▶ Full confusion matrix + precision, recall, F1, accuracy
- ▶ Per-dialogue predictions with explanations saved

Results & Impact

75.0%

Accuracy

88.2%

Precision

83.3%

Recall

85.7%

F1 Score

16.7%

False Neg Rate

Key Takeaways

- ▶ IAP achieves ~30.5% FNR improvement over baselines
- ▶ Intent separation catches subtle manipulation that surface-level analysis misses
- ▶ Detected techniques: gaslighting, guilt-tripping, emotional blackmail, minimization
- ▶ High confidence predictions with explanations enable human-in-the-loop review

Sample Output (per dialogue):

```
{  
  "manipulative": 1,  
  "confidence": 0.9,  
  "technique": "Gaslighting, Guilt-tripping",  
  "explanation": "Person A denies B's ..."  
}
```

New Approach : Memory Over Learning

THE SHIFT



From Learning to Retrieval

- ▶ Moves away from the traditional ML paradigm of fine-tuning weights
- ▶ No classification boundary is learned from data
- ▶ Replaces gradient-based optimization with direct memory lookup

THE MEMORY BANK



Stored Exemplar Database

- ▶ Stores exact, labeled exemplars of both normal and manipulative dialogues
- ▶ Database serves as the model's entire knowledge base
- ▶ Each entry preserves full context and ground-truth label

THE SIGNAL



Semantic Similarity

- ▶ Determines intent purely by measuring semantic similarity
- ▶ No complex neural layers or probability computations
- ▶ Distance in vector space replaces learned decision boundaries

The Retrieval Mechanism

VECTORIZATION



Dense Vector Encoding

- ▶ Converts new conversation segments into dense vector representations
- ▶ Uses embedding models such as sentence-transformers
- ▶ Captures semantic meaning in high-dimensional space

DISTANCE CALCULATION



k-NN Search

- ▶ Executes fast k-nearest-neighbor search in vector space
- ▶ Locates the closest historical examples to the input
- ▶ Optimized algorithms enable sub-millisecond lookups

CLASSIFICATION BY PROXIMITY



Cluster Matching

- ▶ Flags segment as manipulation if it clusters near stored tactics
- ▶ Matches prototypical patterns like "accusation" or "feigning innocence"
- ▶ Semantic proximity drives the final classification decision

Key Advantages & Efficiency

Compute-Light

- ▶ Completely bypasses expensive model training
- ▶ No backpropagation or gradient updates required
- ▶ Runs on minimal hardware — no GPU needed for inference

High Speed

- ▶ Relies on precomputed embeddings for instant lookup
- ▶ Highly optimized nearest-neighbor search algorithms
- ▶ Rapid inference suitable for real-time applications

Granular Output

- ▶ Supports fine-grained taxonomy classification
- ▶ Matches input to specifically labeled clusters
- ▶ Inherent multi-class support without architectural changes

Instant Adaptability

- ▶ Updated in real-time by adding new examples to the vector DB
- ▶ Zero downtime required for system updates
- ▶ New manipulation patterns recognized immediately after insertion

Tradeoffs & Structural Limitations

EMBEDDING DEPENDENCY

Quality Bottleneck

- ▶ Entire system accuracy is bottlenecked by embedding quality
- ▶ If embeddings fail to capture pragmatic nuance, retrieval fails
- ▶ Choice of embedding model is a critical design decision

VULNERABILITY TO NOVELTY

Out-of-Domain Weakness

- ▶ Easily fooled by out-of-domain phrasing patterns
- ▶ Highly novel sentence structures won't map near stored cases
- ▶ Requires continuous exemplar expansion for coverage

SEMANTIC BLIND SPOTS

False Positive Risk

- ▶ Prioritizes semantic closeness over explicit logical reasoning
- ▶ May misclassify benign text sharing vocabulary with manipulative data
- ▶ Lacks ability to reason about context or speaker intent

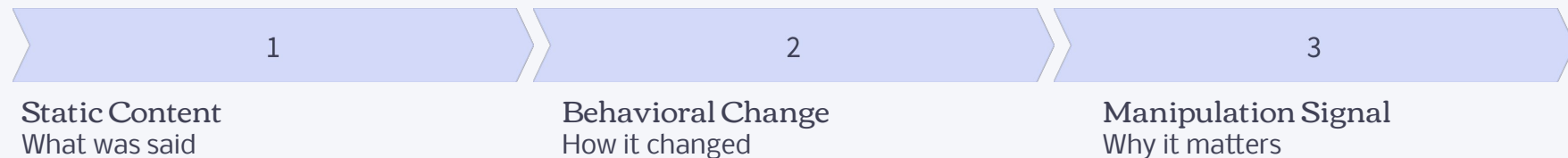
Transition-Based Modeling

The Core Problem

The baseline treats the entire dialogue as a single static object, instead of modeling interactions between turns explicitly

The Reality

Manipulation is reactive, it appears when the system resists or the user changes strategy.



Manipulation is not in the utterance itself. It is in the change between turns.



Tone Shift

Polite → aggressive



Intent Persistence

Rephrased but persistent



Emotion Change

Neutral → negative



Strategy Evolution

Request → pressure → guilt

Transition-Based Detection Pipeline

Five-stage computational flow that detects behavioral change rather than isolated content.



Input Representation

Pair of consecutive turns: U_{t-1} and U_t form transition unit



Independent Encoding

Both turns encoded separately into semantic representations



Transition Construction

Representations compared and combined into difference-aware representation



Signal Extraction

Learned patterns of variation: direction, degree, and consistency of change



Decision Layer

Classification based on interaction between turns, not isolated content

Behavioral Dynamics Captured by the Model



Change-Oriented Representation

The model does not focus on absolute meaning of U_t . Instead it focuses on **relative change from U_{t-1} to U_t**



Reaction Modeling

The system implicitly learns:

- how users behave when **their intent is blocked**
- whether the next turn is:
 - adaptive
 - persistent
 - escalatory

This is modeling **reaction, not language**



Content vs Behavior

- Content = what is being asked
- Behavior = how the user is adjusting

The pipeline prioritizes **behavioral adjustment signals over raw content**

Roles

- **Vinay (MA23BTECH11026) :- (Slides 4-8)**
- **Gopi (MA23BTECH11015) :- (Slides 9-16)**
- **Sattar (MA23BTECH11025) :- (Slide 3, 17-19)**

Thank you